Lesson notes · for you

What is data science?

Data science · Level 1 · Lesson 1

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Say what data science is.
- Follow the loop: ask, collect, explore, explain.
- Turn a vague question into a measurable one.

Activity: Question detectives

1. Warm-up (5 min): Read two questions and decide which one data could answer.
2. Write (10 min): With a partner, write three questions about school and make each one measurable.
3. Count (10 min): Choose one question and collect the answer from your class with tally marks.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- What made a question measurable?
- Where did your number come from?
- What is your next question?

Check yourself

- Write one measurable question about our school, and the record that would answer it.

Try it this week
Write three questions about your school that data could answer. Choose one, count the answer this week, and say where your number came from.

Worksheet

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Name _____ Class _____ Date _____

1 **We explained our finding. What comes next in the loop?**

◯ Stop: we are done ◯ A new question ◯ Delete the data

2 **Put the data detective's steps in order.**

☐ Look at what you found

☐ Ask a question

☐ Share the answer

☐ Collect data

3 **Six pupils voted for their favourite sport: football, chess, football, art, football, chess. Count the votes.**

Football: _____ votes. Chess: _____. Art: _____.

Word bank: 4 · 3 · 2 · 1 · 6

4 **Which is a good data question?**

◯ How many minutes does each of us walk to school? ◯ Is school fun? ◯ Are we good people?

5 **When you answer a data question, you are always finished.**

◯ True ◯ False

✎ **Which question about your school or home would you most like data to answer?**

_____

_____

_____

Check your answers

What is data science?

1 **A new question**

Data science is a loop: every answer leads to the next question, as the Monday pattern led to the bus.

2 **Top to bottom: 3, 1, 4, 2**

First the question, then the data, then looking, then sharing.

3 **Football: 3 votes. Chess: 2. Art: 1.**

Count one for each vote: football 3, chess 2, art 1.

4 **How many minutes does each of us walk to school?**

You can measure minutes, so data can answer it.

5 **False**

An answer usually leads to a new question: data science is a loop.